NOTES ON DERIVED COMPLETION AND DUALITY

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1. Derived torsion and completion

Let A be a ring, let $\underline{f} = (f_1, \dots, f_n)$. Write

$$C^{\underline{f}} = \left[A \longrightarrow \prod_{i=1}^{n} A_{f_i} \longrightarrow \prod_{i_1<i_2} A_{f_{i_1} f_{i_2}} \longrightarrow \cdots \longrightarrow A_{f_1 \cdots f_n} \right] \simeq C^{f_1} \otimes_A^{\mathbf{L}} \cdots \otimes_A^{\mathbf{L}} C^{f_n},$$

where A sits at cohomological degree 0.

Lemma 1.1. *$C^{\underline{f}}$ is represented by a complex of projetive A-modules in the range $[0, n]$.*

Proof. We may assume $n = 1$ and it suffices to show $\mathrm{pd}_A A_f \leq 1$ for all $f \in A$. This is true because we have the explicit resolution

$$0 \longrightarrow \bigoplus_{\mathbf{N}} A \longrightarrow \bigoplus_{\mathbf{N}} A \longrightarrow A_f \longrightarrow 0$$

where the first map sends $(a_0, a_1, \dots)$ to $(a_0, a_1 - fa_0, a_2 - fa_1 \dots)$ and the second map sends $(a_0, a_1, \dots)$ to $a_0 + a_1/f + a_2/f^2 + \dots$. □

There is a distinguished triangle

$$C^{\underline{f}} \longrightarrow A \longrightarrow (A_h\text{-complexes, } h \in (\underline{f})) \longrightarrow +1. \tag{1}$$

For one element f, this is

$$C^f \longrightarrow A \longrightarrow A_f \longrightarrow +1.$$

For $X \in \mathrm{D}(A)$, we have the natural maps

$$R\Gamma_{\underline{f}} X := X \otimes_A^{\mathbf{L}} C^{\underline{f}} \longrightarrow X, \qquad X \longrightarrow \widehat{X}^{\underline{f}} := \mathrm{RHom}_A(C^{\underline{f}}, X).$$

Without any understanding of the notions yet, we already have our first theorem via tensor-hom adjunction.

Theorem 1.2. *For $X, Y \in \mathrm{D}(A)$,*

$$\mathrm{RHom}_A(R\Gamma_{\underline{f}} X, Y) = \mathrm{RHom}_A(X, \widehat{Y}^{\underline{f}}).$$

Definition 1.3. An object $X \in \mathrm{D}(A)$ is *f^∞-torsion* if $X \otimes_A^{\mathbf{L}} A_f = 0$, or equivalently if $R\Gamma_f X \xrightarrow{\sim} X$. It is *$f$-complete* if $\mathrm{RHom}_A(A_f, X) = 0$, or equivalently if $X \xrightarrow{\sim} \widehat{X}^f$. In the torsion condition, one may equivalently require $X \otimes_A^{\mathbf{L}} Z = 0$ for every A_f-complex Z: indeed, this tensor product is $(X \otimes_A^{\mathbf{L}} A_f) \otimes_{A_f}^{\mathbf{L}} Z$. In the completeness condition, one may equivalently require $\mathrm{RHom}_A(Z, X) = 0$ for every A_f-complex Z, by the adjunction

$$\mathrm{RHom}_A(Z, X) = \mathrm{RHom}_{A_f}(Z, \mathrm{RHom}_A(A_f, X)).$$

Theorem 1.4 ([Stacks, Tag 091Q])**.** *Let $X \in \mathrm{D}(A)$. The sets*

$$\{f \in A \mid X \text{ is } f^\infty\text{-torsion}\}, \qquad \{f \in A \mid X \text{ is } f\text{-complete}\}$$

are radical ideals.

If these ideals contain an ideal I, we say that X is I^∞-*torsion* and I-*complete*, respectively.

Proof. To see the sets are closed under sums, we have a distinguished triangle

$$A_{f+g} \longrightarrow A_{f(f+g)} \oplus A_{g(f+g)} \longrightarrow A_{fg(f+g)} \longrightarrow +1.$$

Indeed, in A_{f+g} the elements f and g generate the unit ideal. Tensoring this triangle with X proves closure under addition for the torsion condition. Applying $\mathrm{RHom}_A(-, X)$ proves closure under addition for the completeness condition. Similarly but more easily, we get that the sets are closed under products. □

Theorem 1.5. *The inclusions of the thick full subcategories*

$$\mathrm{D}_{I\text{-comp}}(A) \subseteq \mathrm{D}(A) \supseteq \mathrm{D}_{I^\infty\text{-tors}}(A)$$

have adjoints

$$\widehat{(-)}^{\underline{f}} \dashv (\mathrm{D}_{I\text{-comp}}(A) \hookrightarrow \mathrm{D}(A)), \qquad (\mathrm{D}_{I^\infty\text{-tors}}(A) \hookrightarrow \mathrm{D}(A)) \dashv R\Gamma_{\underline{f}}(-).$$

In particular, these functors do not depend on $\underline{f}$*, but only on* $\sqrt{I}$*, where* $I = (\underline{f})$*. We will write* $\widehat{(-)}^I$ *and* $R\Gamma_I(-)$ *instead.*

We use

Lemma 1.6. *For every* A_f*-complex* X*,* $\mathrm{RHom}_A(A_f, X) = X$*; alternatively,* $\widehat{X}^f = 0$*.*

Proof. Since $\mathrm{pd}_A A_f \leq 1$, we may assume $X \in \mathrm{D}^+(A)$. Then X is represented by a complex of injective A-modules that are also A_f-modules: the injective hull of every f^∞-torsion-free module is f^∞-torsion-free, hence an A_f-module. Now the result follows from $\mathrm{Hom}_A(A_f, M) = M$ for every $M \in \mathrm{D}^\heartsuit(A_f)$. □

Proof. We only need to show $C^{\underline{f}} \otimes^{\mathbf{L}}_A A_{f_i} = 0$ for every i. This follows from $[A \longrightarrow A_{f_i}] \otimes^{\mathbf{L}}_A A_{f_i} = 0$. □

Proof of Theorem 1.5. Let $Y \in \mathrm{D}_{I\text{-comp}}(A)$ and $X \in \mathrm{D}(A)$. The map $X \to \widehat{X}$ induces

$$\mathrm{RHom}_A(\widehat{X}, Y) \longrightarrow \mathrm{RHom}_A(X, Y).$$

Its fiber is of the form $\mathrm{RHom}_A(A_f\text{-complexes}, Y)$ where $f \in I$ by (1), hence zero.

Let $X \in \mathrm{D}_{I^\infty\text{-tors}}(A)$ and $Y \in \mathrm{D}(A)$. The natural map induces

$$\mathrm{RHom}_A(X, R\Gamma_{\underline{f}} Y) \longrightarrow \mathrm{RHom}_A(X, Y).$$

Its cofiber is of the form $\mathrm{RHom}_A(X, A_f\text{-complexes})$ where $f \in I$ by (1). For $f \in I$ and Z an A_f-complexes, by Lemma 1.6 and adjunction,

$$\begin{aligned}\mathrm{RHom}_A(X, Z) &= \mathrm{RHom}_A(X, \mathrm{RHom}_A(A_f, Z)) \\ &= \mathrm{RHom}_A(X \otimes^{\mathbf{L}}_A A_f, Z) = 0,\end{aligned}$$

as $X \otimes^{\mathbf{L}}_A A_f = 0$ □

Lemma 1.7. *The canonical map*

$$C^{\underline{f}} \otimes^{\mathbf{L}}_A C^{\underline{f}} \xrightarrow{\sim} C^{\underline{f}}$$

is an isomorphism.

Proof. Looking at the definition of $C^{\underline{f}}$ we need to show for all i and all $h \in A$ we have $C^{\underline{f}} \otimes^{\mathbf{L}}_A A_{f_i h} = 0$. This is because $C^{f_i} \otimes^{\mathbf{L}}_A A_{f_i h} = [A_{f_i h} \to A_{f_i f_i h}] = 0$. □

Corollary 1.8. *The functors* $\widehat{(-)}^I$ *and* $R\Gamma_I(-)$ *are idempotent.*

Lemma 1.9. *Omitting the ideal from the completion notation, we have*

$$\widehat{\mathrm{RHom}_A(X,Y)} = \mathrm{RHom}_A(\widehat{X},\widehat{Y}) = \mathrm{RHom}_A(X,\widehat{Y}).$$

Proof. The second equality follows from Theorem 1.5. For the first,

$$\begin{aligned}\mathrm{RHom}_A(C^{\underline{f}}, \mathrm{RHom}_A(X,Y)) &= \mathrm{RHom}_A(C^{\underline{f}} \otimes^{\mathbf{L}}_A X, Y)\\ &= \mathrm{RHom}_A(X \otimes^{\mathbf{L}}_A C^{\underline{f}}, Y)\\ &= \mathrm{RHom}_A(X, \widehat{Y}),\end{aligned}$$

by Theorem 1.2. □

Theorem 1.10.

$$\mathrm{RHom}_A(\widehat{X}^I, \widehat{Y}^I) = \mathrm{RHom}_A(R\Gamma_I X, R\Gamma_I Y).$$

Proof. By Theorem 1.2, the right side is $\mathrm{RHom}_A(X, \widehat{R\Gamma_I Y})$, whereas the left side is $\mathrm{RHom}_A(X,\widehat{Y})$ by Lemma 1.9. Consider the triangle

$$R\Gamma_I Y \longrightarrow Y \longrightarrow Q \longrightarrow +1.$$

We need $\widehat{Q} = 0$. It suffices to show that $\widehat{Z} = 0$ for every A_f-complex Z, where $f \in I$. This is because we can use any set of generators $f = f_1, \dots, f_n$ of I to compute $\widehat{Z}$, so $\widehat{Z} = \mathrm{RHom}_A(C^{\underline{f}}, Z) = \mathrm{RHom}_A(C^{f_2,\dots,f_n}, \mathrm{RHom}_A(C^f, Z)) = 0$ (Lemma 1.6). □

Corollary 1.11. *For every $Y \in \mathrm{D}(A)$, the natural map induces an isomorphism*

$$\widehat{R\Gamma_{\underline{f}} Y} \xrightarrow{\sim} \widehat{Y}.$$

Proof. In the triangle in the preceding proof, $\widehat{Q} = 0$. Applying completion therefore gives the asserted isomorphism. □

Proposition 1.12. *For every $X \in \mathrm{D}(A)$,*

$$\begin{aligned} X \in \mathrm{D}_{\mathrm{tors}}(A) &\iff \mathrm{H}^i(X) \in \mathrm{D}^{\heartsuit}_{\mathrm{tors}}(A) \text{ for every } i,\\ X \in \mathrm{D}_{\mathrm{comp}}(A) &\iff \mathrm{H}^i(X) \in \mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A) \text{ for every } i.\end{aligned}$$

Proof. The first assertion is immediate because tensoring with A_f is flat. For the second, $\mathrm{RHom}_A(A_f, -)$ has cohomological dimension $[-1, 0]$, so we have

$$\begin{aligned}&\mathrm{Ext}^1_A(A_f, \mathrm{H}^{i-1}(X)) = 0 = \mathrm{Hom}_A(A_f, \mathrm{H}^i(X))\\ &\iff \mathrm{H}^i \mathrm{RHom}_A(A_f, X) = 0.\end{aligned}$$

□

Corollary 1.13. $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ *is a weak Serre subcategory.*

Proof. As an illustration, let us see why $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ is stable under kernels and cokernels. Let $\varphi : M \to N \in \mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$, then $[M \to N] \in \mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ as $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ is a thick subcategory. Apply Proposition 1.12 we see $\ker(\varphi)$ and $\mathrm{coker}(\varphi)$ are in $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$. □

Lemma 1.14 (Nakayama)**.** *Let $M \in \mathrm{D}^{\heartsuit}_{I\text{-comp}}(A)$. If $M = IM$, then $M = 0$.*

Proof. Say $I = (f_1, \dots, f_n)$. By Corollary 1.13, $N = (f_2, \dots, f_n)M$ and M/N are in $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ (N being the image of a map $M^{\oplus(n-1)} \to M$). We have $M = f_1 M + N$. Induction on n reduces us to $n = 1$. Then $M = fM$, and, if $M \neq 0$, elementary considerations give $\mathrm{Hom}_A(A_f, M) \neq 0$. □

Corollary 1.15. *Let $X \in \mathrm{D}_{I\text{-comp}}(A)$ where $I = (f_1, \dots, f_n)$. The following are equivalent for every integer c.*

(*i*) $X \in \mathrm{D}^{\leq c}$.
(*ii*) $X \otimes^{\mathbf{L}}_A A/I \in \mathrm{D}^{\leq c}$.
(*iii*) $X \otimes^{\mathbf{L}}_A K \in \mathrm{D}^{\leq c}$, *where* $K = \mathrm{Kos}(f_1, \ldots, f_n)$ *is the Koszul complex.*

Proof. Trivially (*i*) implies (*ii*), and (*ii*) implies (*iii*) as $X \otimes^{\mathbf{L}}_A K = X \otimes^{\mathbf{L}}_A A/I \otimes^{\mathbf{L}}_{A/I} K$. Now assume (*iii*) and let $Y = X \otimes^{\mathbf{L}}_A \mathrm{Kos}(f_1, \ldots, f_{n-1})$. As $\mathrm{Kos}(f_1, \ldots, f_{n-1})$ is a perfect complex we have $Y \in \mathrm{D}_{I\text{-comp}}(A)$, and (*iii*) gives $\mathrm{H}^i(Y) = f_n \mathrm{H}^i(Y)$ for all $i > 0$. Proposition 1.12 and Lemma 1.14 give $\mathrm{H}^i(Y) = 0$ for all $i > 0$, and we win by induction. □

Lemma 1.16 ([Lyu25, Lemma 5.4])**.** *Let* $X \in \mathrm{D}_{I\text{-comp}}(A)$, $f \in I$, *and*

$$C := \mathrm{cofib}(X \xrightarrow{\times f} X).$$

(*i*) *If* $C = 0$, *then* $X = 0$.
(*ii*) *Assume* A *is Noetherian. If* $A \in \mathrm{D}_{I\text{-comp}}(A)$ *and* $X \in \mathrm{D}^-(A)$, *then* $C \in \mathrm{D}^-_{\mathrm{coh}}(A)$ *implies* $X \in \mathrm{D}^-_{\mathrm{coh}}(A)$.

Proof. We get (*i*) trivially from Corollary 1.15. To prove (*ii*), fix $c \in \mathbf{Z}$ and let us show $\mathrm{H}^c(X)$ is finitely generated.

Let $b \in \mathbf{Z}$ be such that $X \in \mathrm{D}^{\leq b}$ and not in $\mathrm{D}^{\leq b-1}$. We use induction on b, as when $b < c$ the result is trivial. We have $\mathrm{H}^b(X) \in \mathrm{D}_{I\text{-comp}}(A)$ and $\mathrm{H}^b(X)/f\mathrm{H}^b(X)$ is finitely generated. Therefore there exists a map $\varphi : F \to \mathrm{H}^b(X)$ inducing a surjective map on $(-)/f(-)$, where F is a finite free A-module. Then φ is surjective by Lemma 1.14. The map φ comes from a map $\psi : F[-b] \to X$, and as $A \in \mathrm{D}_{I\text{-comp}}(A)$, we have $\mathrm{cofib}(\psi) \in \mathrm{D}_{I\text{-comp}}(A)$, which has smaller b, we win. □

2. Finitely genereted modules over Noetherian rings

Theorem 2.1 (Derived and classical completions coincide for finite modules over Noetherian rings)**.** *Let* A *be a Noetherian ring and* I *and ideal of* A. *Then for every* $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$ *we have* $\widehat{M}^I = \lim_t M/I^t M$.

[To be built in the future. This follows from [Stacks, Tag 0A06]; we want to supplement an interesting elementary argument that avoids derived limits.]

3. Matlis duality

In the remainder of this note, let $(A, \mathfrak{m}, k)$ be a Noetherian complete local ring (in particular, $\mathfrak{m}$ is finitely generated). We omit $\mathfrak{m}$ from the notation when it is obvious (e.g. $\mathrm{D}_{\mathrm{comp}}$ means $\mathrm{D}_{\mathfrak{m}\text{-comp}}$).

We give the axiomatic description of the injective hull of k (cf. [Stacks, Tag 08Y3]). We will see such an object is unique (Prop 5.2).

Theorem 3.1. *There exists* $E \in \mathrm{D}^{\heartsuit}(A)$ *such that*

(*i*) E *is injective;*
(*ii*) $\mathrm{Hom}_A(k, E) \simeq k$;
(*iii*) E *is* $\mathfrak{m}^\infty$*-torsion (cf. [Stacks, Tag 08XW]).*

Note that (*i*) and (*ii*) imply $\mathrm{RHom}_A(k, E) \simeq k$.

Define

$$(-)^\vee := \mathrm{RHom}_A(-, E)$$

so for all $M \in \mathrm{D}(A)$ we have a canonical map $\mu_M : M \longrightarrow M^{\vee\vee}$.

Remark 3.2. The canonical map $\mu_k : k \to k^{\vee\vee}$ is an isomorphism. Indeed, $\mu_M : M \longrightarrow M^{\vee\vee}$ is nonzero for any $M \in \mathrm{D}^{\heartsuit}(A)$ as soon as $\mathrm{RHom}_A(M, E) \neq 0$, since $\mu_M(x) = (\varphi \mapsto \varphi(x))$ is the evaluation map. But any nonzero map between two isomorphic copies of k is an isomorphism.

Theorem 3.3. *For any $M \in \mathrm{D}_{\mathrm{coh}}(A)$, the canonical map*

$$\mu_M : M \longrightarrow \mathrm{RHom}_A(\mathrm{RHom}_A(M, E), E)$$

is an isomorphism.

Proof. Since E is injective, we may assume M is bounded. Induct on $d = \dim \mathrm{Supp}\, M$. If $d = 0$, M is a successive extension of shifts of k, so we win by Remark 3.2. Assume $d > 0$ and the theorem is true fo smaller d, take $f \in \mathfrak{m}$ such that

$$\dim(V(f) \cap \mathrm{Supp}\, M) < d.$$

Consider the distinguished triangle which defines C

$$M \xrightarrow{\times f} M \longrightarrow C \longrightarrow +1.$$

Applying double duality gives a morphism of distinguished triangles

$$\begin{array}{ccccccc} M & \xrightarrow{\times f} & M & \longrightarrow & C & \longrightarrow & +1 \\ \downarrow{\scriptstyle \mu_M} & & \downarrow{\scriptstyle \mu_M} & & \downarrow{\scriptstyle \mu_C} & & \\ M^{\vee\vee} & \xrightarrow{\times f} & M^{\vee\vee} & \longrightarrow & C^{\vee\vee} & \longrightarrow & +1 \end{array}$$

We know $\mathrm{Supp}\, C \subseteq V(f) \cap \mathrm{Supp}\, M$. By the induction hypothesis, μ_C is an isomorphism. Thus $\times f$ is an isomorphism on $\mathrm{cofib}(\mu_M)$. However, $M \in \mathrm{D}_{\mathrm{comp}}(A)$ as A is complete, and $M^{\vee\vee} \in \mathrm{D}_{\mathrm{comp}}(A)$: both $\mathrm{RHom}_A(M, E)$ and E are $\mathfrak{m}^\infty$-torsion, as $M \in \mathrm{D}^b_{\mathrm{coh}}(A)$, so Theorem **??** and Lemma **??** tell us $M^{\vee\vee} \in \mathrm{D}_{\mathrm{comp}}(A)$. Hence $\mathrm{cofib}(\mu_M) \in \mathrm{D}_{\mathrm{comp}}(A)$. By Nakayama, $\mathrm{cofib}(\mu_M) = 0$. □

Corollary 3.4. $\mathrm{Hom}_A(E, E) = A$.

Remark 3.5. There is an equivalence of abelian categories (Matlis duality)

$$\{\text{Noetherian (i.e. finitely generated) } A\text{-modules}\} \underset{(-)^\vee}{\overset{(-)^\vee}{\rightleftarrows}} \{\text{Artinian } A\text{-modules}\}.$$

We have proved part of it ($\mathrm{id} = (-)^{\vee\vee}$ on the category on the left hand side). The rest of the proof is left to the interested reader (not trivial).

4. Local duality

Definition 4.1. The normalized dualizing complex is

$$\omega_A^\bullet := \widehat{E},$$

and we set, for every $M \in \mathrm{D}(A)$

$$\mathbb{D}(M) := \mathrm{RHom}_A(M, \omega_A^\bullet) \in \mathrm{D}_{\mathrm{comp}}(A) \text{ (cf. Lemma 1.9)}.$$

Again uniqueness will be dealt with the next time.

Theorem 4.2 (The local duality theorem, [Stacks, Tag 0A84])**.** *For $M \in \mathrm{D}(A)$,*

$$\mathbb{D}(M) = \mathrm{RHom}_A(R\Gamma_{\mathfrak{m}} M, E) \in \mathrm{D}_{\mathrm{comp}}(A).$$

In particular, $\mathbb{D}(k) \simeq k$.

Corollary 4.3. *If $M \in \mathrm{D}^{[a,b]}(A)$, then $\mathbb{D}(M) \in \mathrm{D}^{[-b-\dim A, -a]}(A)$. If $M \in \mathrm{D}^{\heartsuit}(A)$, then $\mathbb{D}(M) \in \mathrm{D}^{[-\dim A, 0]}(A)$. (This is because $\mathfrak{m}$ is the radical of an ideal generated by $\dim A$ elements, hence $R\Gamma_{\mathfrak{m}}(-)$ has cohomological dimension $\dim A$.)*

In particular, $\omega_A^\bullet \in \mathrm{D}^{[-\dim A, 0]}(A)$.

Proof of Theorem 4.2.

$$\begin{aligned}
\mathbb{D}(M) &= \mathrm{RHom}_A(M, \omega_A^\bullet) \\
&= \mathrm{RHom}_A(\widehat{M}, \widehat{E}) && \text{(Lemma 1.9 and Definition)} \\
&= \mathrm{RHom}_A(R\Gamma_{\mathfrak{m}} M, R\Gamma_{\mathfrak{m}} E) && \text{(Theorem 1.10)} \\
&= \mathrm{RHom}_A(R\Gamma_{\mathfrak{m}} M, E) && (E \in \mathrm{D}_{\mathrm{tors}}(A)).
\end{aligned}$$

□

Theorem 4.4. *For every $M \in \mathrm{D}_{\mathrm{coh}}(A)$, the canonical map*

$$\gamma_M : M \longrightarrow \mathbb{D}(\mathbb{D}(M))$$

is an isomorphism.

Proof. The proof is the same as for μ_M. Indeed, by Corollary 4.3, we may assume $M \in \mathrm{D}^b_{\mathrm{coh}}(A)$. Using $\mathbb{D}(\mathbb{D}(M)) \in \mathrm{D}_{\mathrm{comp}}(A)$, we can reduce to the case $M = k$. Then Theorem **??** tells us for $X \in \mathrm{D}_{\mathrm{comp}} \cap \mathrm{D}_{\mathrm{tors}}$ we have $\mathbb{D}(X) = X^\vee$. Therefore

$$\mathbb{D}(\mathbb{D}(k)) = \mathbb{D}(k^\vee) = k^{\vee\vee},$$

and γ_k corresponds to μ_k. □

Theorem 4.5. *If $M \in \mathrm{D}_{\mathrm{coh}}(A)$, then $\mathbb{D}(M) \in \mathrm{D}_{\mathrm{coh}}(A)$ (in particular, $\omega_A^\bullet \in \mathrm{D}_{\mathrm{coh}}(A)$).*

Proof. By Corollary 4.3, it suffices to consider the case $M \in \mathrm{D}^b_{\mathrm{coh}}(A)$. Again, induct on $d = \dim \operatorname{Supp} M$. If $d = 0$, we may assume $M = k$, and $\mathbb{D}(k) \simeq k$. If $d > 0$, take $f \in \mathfrak{m}$ with $\dim(V(f) \cap \operatorname{Supp} M) < d$ and consider the triangle

$$M \xrightarrow{\times f} M \longrightarrow C \longrightarrow +1$$

which defines C. Dualizing gives

$$\mathbb{D}(C) \longrightarrow \mathbb{D}(M) \xrightarrow{\times f} \mathbb{D}(M) \longrightarrow +1.$$

By induction, $\mathbb{D}(C) \in \mathrm{D}^b_{\mathrm{coh}}(A)$, and $\mathbb{D}(M) \in \mathrm{D}^b_{\mathrm{comp}}(A)$ (Theorem 4.2 and Corollary 4.3). By Nakayama, $\mathbb{D}(M) \in \mathrm{D}_{\mathrm{coh}}(A)$. □

5. Uniqueness

Lemma 5.1. *Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$. If*

$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\geq \alpha}(A),$$

then $X \in \mathrm{D}^{\geq \alpha}(A)$.

Proof. Let β be such that $X \in \mathrm{D}^{\geq \beta}$ and $X \notin \mathrm{D}^{\geq \beta+1}$. Then

$$\mathrm{H}^\beta \mathrm{RHom}_A(k, X) \simeq \mathrm{Hom}_A(k, \mathrm{H}^\beta(X)) \neq 0.$$

Thus $\beta \geq \alpha$. □

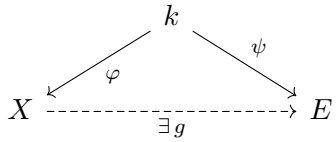
The more general form of Lemma 5.1 is dual to Corollary 1.15 and is left as an exercise.

Proposition 5.2. *Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$ be such that $\mathrm{RHom}_A(k, X) \simeq k$. Then $X \simeq E$.*

Proof. The proof proceeds as follows.

(1) $X \in \mathrm{D}^{\geq 0}$ (Lemma 5.1).

(2) Let $\varphi : k \to X$ be a generator of $\mathrm{Hom}_{\mathrm{D}(A)}(k, X) = \mathrm{H}^0(\mathrm{RHom}_A(k, X)) \simeq k$, and $\psi : k \to E$ be a generator of $\mathrm{Hom}_{\mathrm{D}(A)}(k, E) = \mathrm{H}^0(\mathrm{RHom}_A(k, E)) \simeq k$, Set $C = \mathrm{cofib}(\varphi)$. Then $C \in \mathrm{D}^{\geq 0}(A)$, so $\mathrm{RHom}_A(C, E) \in \mathrm{D}^{\leq 0}(A)$, therefore in the following diagram g exists.



(3) By choice, both φ and ψ induce an isomorphism on $\mathrm{Hom}_{\mathrm{D}(A)}(k, -)$, thus by assumption, g induces an isomorphism on $\mathrm{RHom}_A(k, -)$. Hence

$$\mathrm{RHom}_A(k, \mathrm{cofib}(g)) = 0,$$

which implies $\mathrm{cofib}(g) = 0$ (Lemma 5.1).

Consequently $X \simeq E$. □

Proposition 5.3. *Let $Y \in \mathrm{D}^+_{\mathrm{comp}}(A)$ be such that $\mathrm{RHom}_A(k, Y) \simeq k$. Then $Y \simeq \omega^\bullet_A$.*

Proof. We have

$$\mathrm{RHom}_A(k, \mathrm{R}\,\Gamma_{\mathfrak{m}} Y) \simeq k,$$

so $\mathrm{R}\,\Gamma_{\mathfrak{m}} Y \simeq E$ by the preceding proposition. Hence

$$Y \simeq \widehat{\mathrm{R}\,\Gamma_{\mathfrak{m}} Y} \simeq \widehat{E} = \omega^\bullet_A.$$

□

6. Dualizing complexes: the standard definition

Definition 6.1. Let A be an arbitrary Noetherian ring. An object $\omega^\bullet \in \mathrm{D}(A)$ is a *dualizing complex* if

(1) $\omega^\bullet$ has finite injective dimension, i.e. there exists $\alpha \in \mathbb{Z}$ such that for every $X \in \mathrm{D}^\heartsuit(A)$,

$$\mathrm{RHom}_A(X, \omega^\bullet) \in \mathrm{D}^{\leq \alpha}(A);$$

this is also equivalent to that $\omega^\bullet$ is represented by a bounded complex of injectives.

(2) $\omega^\bullet \in \mathrm{D}_{\mathrm{coh}}(A)$.

(3) the natural map is an isomorphism:

$$A = \mathrm{RHom}_A(\omega^\bullet, \omega^\bullet).$$

Note that dualizing complexes are stable under localization [Stacks, Tag 0A7G].

Theorem 6.2. *If A is local, then*

$$\omega^\bullet \otimes^L_A \widehat{A} \simeq \omega^\bullet_{\widehat{A}}[\delta]$$

for some $\delta \in \mathbf{Z}$.

We say $\omega^\bullet$ is *normalized* if $\delta = 0$.

Proof. It suffices to show (by Prop 5.3 and the fact that RHom commutes with flat base change if the source in pseudo-coherent)

$$\mathrm{RHom}_A(k, \omega^\bullet) \simeq k[\delta].$$

From conditions (1) and (3), for $M \in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(A)$,

$$M = \mathrm{RHom}_A(\mathrm{RHom}_A(M, \omega^\bullet), \omega^\bullet).$$

In particular,

$$k = \mathrm{RHom}_A(\mathrm{RHom}_A(k, \omega^\bullet), \omega^\bullet).$$

By (1) and (2), $\mathrm{RHom}_A(k,\omega^\bullet)\in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(k)$, hence is of the form $\bigoplus_{i=1}^t k[a_i]^{\oplus d_i}$. Then

$$k=\mathrm{RHom}_A(\mathrm{RHom}_A(k,\omega^\bullet),\omega^\bullet)\cong\bigoplus_{j=1}^t\left(\bigoplus_{i=1}^t k[a_i]^{\oplus d_i}\right)[a_j]^{\oplus d_j},.$$

forcing $t=1=d_1$; and we put $\delta=a_1$. □

Remark 6.3. Not all Noetherian rings have dualizing complexes. If a local ring has two, they are isomorphic up to a shift. This is because if $X,Y\in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(A)$ are such that $X\otimes^L_A \widehat{A}\simeq Y\otimes^L_A\widehat{A}$, then $X\simeq Y$. This is because $\mathrm{Hom}_{\mathrm{D}(A)}(X,Y)$ is dence in its completion and because

$$\{\text{quasi-isomorphisms}\}+\mathfrak{m}\,\mathrm{Hom}_{\mathrm{D}(\widehat{A})}(X\otimes^L_A\widehat{A},Y\otimes^L_A\widehat{A})\subseteq\{\text{quasi-isomorphisms}\}$$

by Nakayama's Lemma.

7. Properties of dualizing complexes

Lemma 7.1. *Let $A\to B$ be a finite ring map, and let $\omega^\bullet$ be a dualizing complex of A. Then*

$$\omega^\bullet_B:=\mathrm{RHom}_A(B,\omega^\bullet)$$

is a dualizing complex of B.

If both A and B are local and $\omega^\bullet$ is normalized, then $\omega^\bullet_B$ is normalized.

Proof. For $N\in\mathrm{D}(B)$, adjunction gives

$$\mathrm{RHom}_B(N,\mathrm{RHom}_A(B,\omega^\bullet))=\mathrm{RHom}_A(N,\omega^\bullet).$$

Thus properties (1) and (2) in Definition 6.1 follow. For property (3),

$$\mathrm{RHom}_B(\omega^\bullet_B,\omega^\bullet_B)=\mathrm{RHom}_A(\omega^\bullet_B,\omega^\bullet)=B,$$

by biduality for the finite A-module B.

For the normalized statement, we have

$$\mathrm{RHom}_B(l,\omega^\bullet_B)=\mathrm{RHom}_A(l,\omega^\bullet),$$

where l is the residue field of B, which is isomorphic to a finite direct sum of k as an A-module. □

8. Dimension function

For a normalized dualizing complex $\omega^\bullet$, define $\delta(\mathfrak{p})$ to be the unique integer d such that

$$\mathrm{H}^d(k(\mathfrak{p}),\omega^\bullet_{\mathfrak{p}})\neq 0.$$

If A is local then $\omega^\bullet$ is normalized if and only if $\delta(\mathfrak{m})=0$.

Theorem 8.1. *If $\mathfrak{p}\subseteq\mathfrak{q}$ and $\mathrm{ht}(\mathfrak{q}/\mathfrak{p})=1$, then*

$$\delta(\mathfrak{p})=\delta(\mathfrak{q})+1.$$

This means that δ is a dimension function.

Proof. By localization and quotient (Lemma 7.1) we may assume $(A,\mathfrak{m})$ is local, $\mathfrak{q}=\mathfrak{m}$, and $\mathfrak{p}=0$, so $\dim A=1$. We may also assume $\omega^\bullet$ is normalized. Then

$$\widehat{\mathrm{H}^{-1}(\omega^\bullet)}\simeq\mathrm{H}^1_{\mathfrak{m}}(A)^\vee,\qquad \widehat{\mathrm{H}^0(\omega^\bullet)}\simeq\mathrm{H}^0_{\mathfrak{m}}(A)^\vee.$$

Since $\mathrm{H}^0_{\mathfrak{m}}(A)$ is of finite length (being an ideal of A)[1], the finitely generated module $\mathrm{H}^0(\omega^\bullet)$ is also of finite length. Since $\mathrm{Supp}(\omega^\bullet)=\mathrm{Spec}\,A$ we must have $\mathrm{Supp}\,\mathrm{H}^{-1}(\omega^\bullet)=\mathrm{Spec}\,A$, whence by definition $\delta(\mathfrak{p})=1$. □

[1] In fact, since we assumed A is a domain, we have $\mathrm{H}^0_{\mathfrak{m}}(A)=0$.

Corollary 8.2. *If a Noetherian ring A has a dualizing complex then A is catenary,* i.e.

$$\operatorname{ht}(\mathfrak{r}/\mathfrak{q}) + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}) = \operatorname{ht}(\mathfrak{r}/\mathfrak{p})$$

for primes $\mathfrak{r} \supseteq \mathfrak{q} \supseteq \mathfrak{p}$.

9. Support of the dualizing module

Theorem 9.1. *Suppose A is local and let $\omega_A^\bullet$ be a normalized dualizing complex (if exists). Let $d = \dim A$. Then*

$$\operatorname{Ass}_A \mathrm{H}^{-d}(\omega_A^\bullet) = \{\mathfrak{p} \mid \dim A/\mathfrak{p} = d\}.$$

Proof. For every $\mathfrak{p} \in \operatorname{Spec} A$, $\operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet)$ is a normalized dualizing complex of $A/\mathfrak{p}$, hence is in $\mathrm{D}^{[-\dim A/\mathfrak{p}, 0]} \subseteq \mathrm{D}^{[-d,0]}$ (Lemma 7.1). If there is an injection

$$A/\mathfrak{p} \hookrightarrow \mathrm{H}^{-d}(\omega_A^\bullet),$$

then

$$\mathrm{H}^{-d} \operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet) = \operatorname{Hom}_A(A/\mathfrak{p}, \mathrm{H}^{-d}(\omega_A^\bullet)) \neq 0.$$

It follows that $\dim A/\mathfrak{p} \geq d$.

Conversely, if $\dim A/\mathfrak{p} = d$, then $\delta(\mathfrak{p}) = -d$ (Theorem 8.1), that is,

$$\big(\mathrm{H}^{-d} \operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet)\big)_\mathfrak{p} \neq 0.$$

Hence there is a nonzero map

$$k(\mathfrak{p}) \longrightarrow \mathrm{H}^{-d}(\omega_A^\bullet)_\mathfrak{p},$$

which gives an injection $A/\mathfrak{p} \hookrightarrow \mathrm{H}^{-d}(\omega_A^\bullet)$ that localizes to the given map. □

Corollary 9.2.

$$\operatorname{Supp} \mathrm{H}^{-d}(\omega_A^\bullet) = \bigcup_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \dim A/\mathfrak{p} = d}} V(\mathfrak{p}).$$

The following statements are left as exercises.

$$\operatorname{Ass}_A \mathrm{H}^{-j}(\omega_A^\bullet) \supseteq \{\mathfrak{p} \in \operatorname{Min} A \mid \dim A/\mathfrak{p} = j\} \tag{2}$$

$$\dim \operatorname{Supp} \mathrm{H}^{-j}(\omega_A^\bullet) \leq j \tag{3}$$

Here $\operatorname{Min} A$ is the set of minimal primes of A.

References

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